

# DVF QA metrics — phase 10

| Metric                       | Value       |
|------------------------------|-------------|
| qa_status                    | FAIL        |
| jac_min                      | -0.815854   |
| jac_p01                      | 0.596034    |
| jac_p05                      | 0.746188    |
| jac_median                   | 0.991718    |
| jac_mean                     | 0.998473    |
| jac_p95                      | 1.27917     |
| jac_p99                      | 1.45635     |
| jac_max                      | 2.87693     |
| fraction_jac_le_0            | 0.00017184  |
| fraction_jac_lt_0_2          | 0.000623872 |
| fraction_jac_gt_5            | 0           |
| folding_component_count      | 1738        |
| largest_folding_component_ml | 0.012       |
| dvf_magnitude_p95_mm         | 4.70766     |
| dvf_magnitude_max_mm         | 10.2439     |
| gradient_norm_mean           | 0.257344    |
| bending_energy_mean          | 0.0580262   |
| warped_vs_fixed_mse          | 8711.88     |
| warped_vs_fixed_mae          | 30.6809     |
| warped_vs_fixed_ncc          | 0.981556    |
| our_warp_vs_precomputed_mse  | 8711.88     |
| our_warp_vs_precomputed_mae  | 30.6809     |
| our_warp_vs_precomputed_ncc  | 0.981556    |

CT comparison (moving / fixed / warped / |diff|)

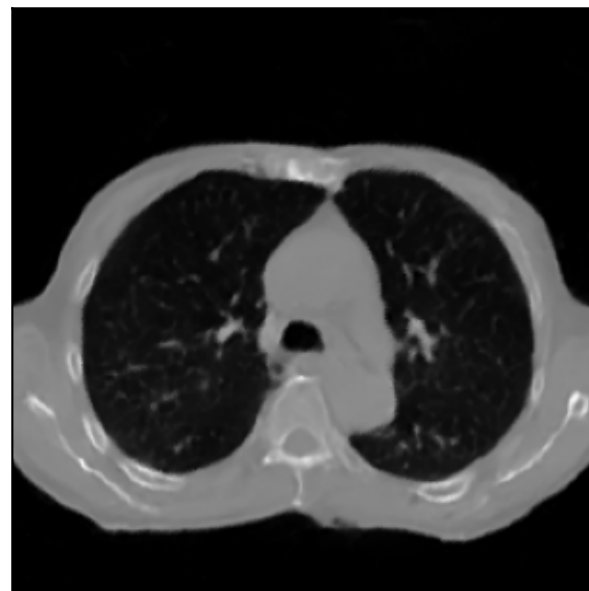
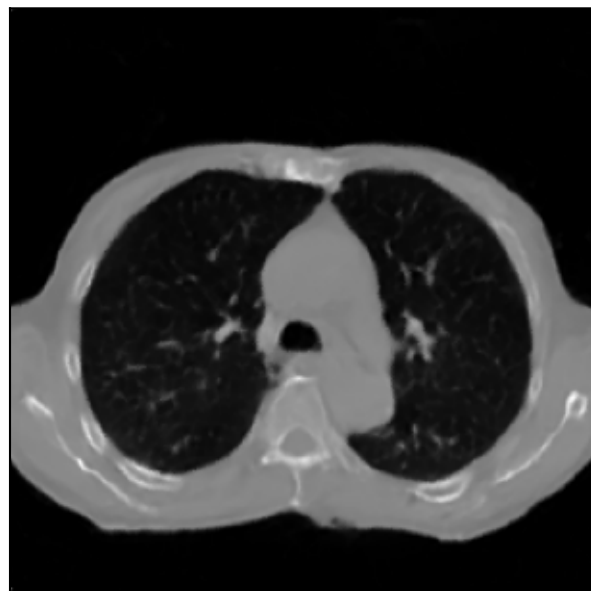
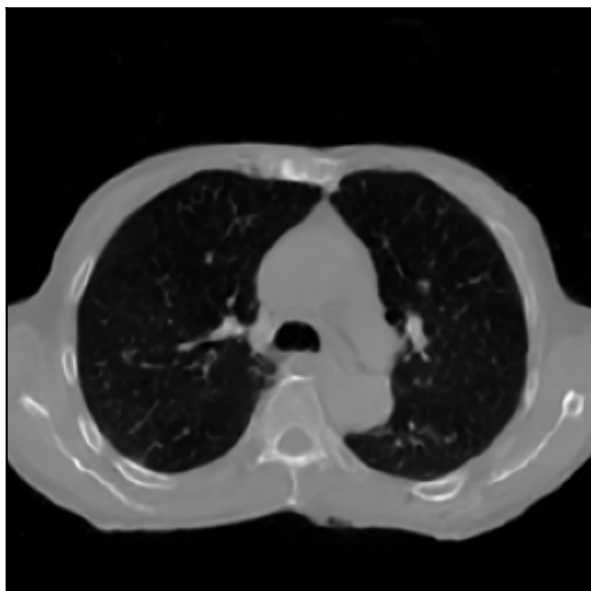
Moving (phase 50)

Fixed (phase 10)

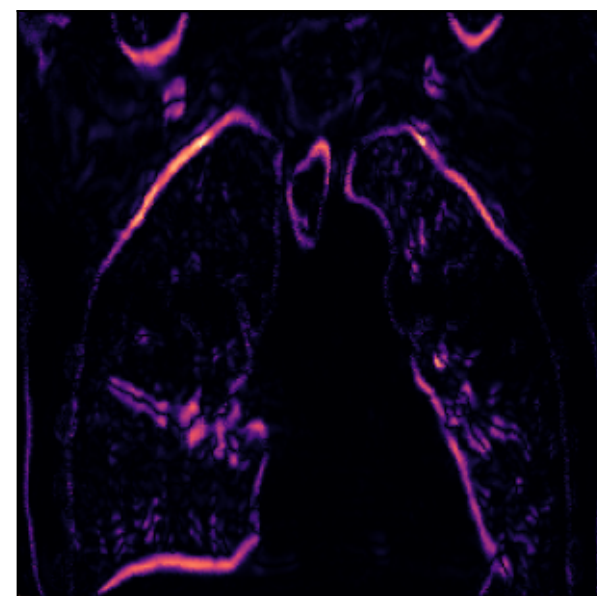
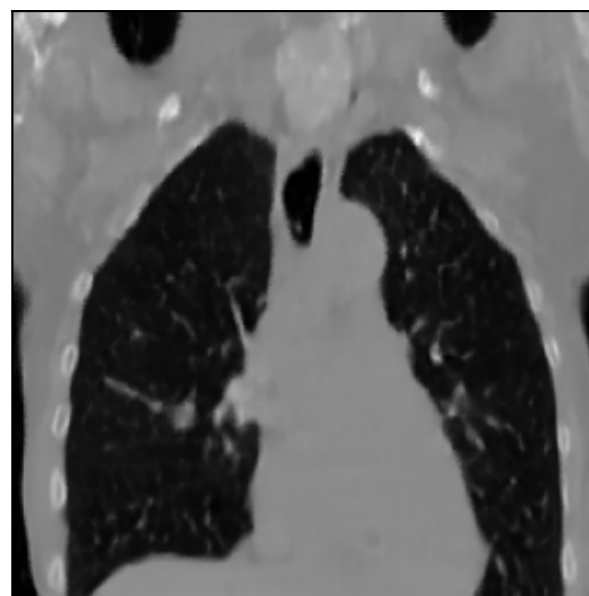
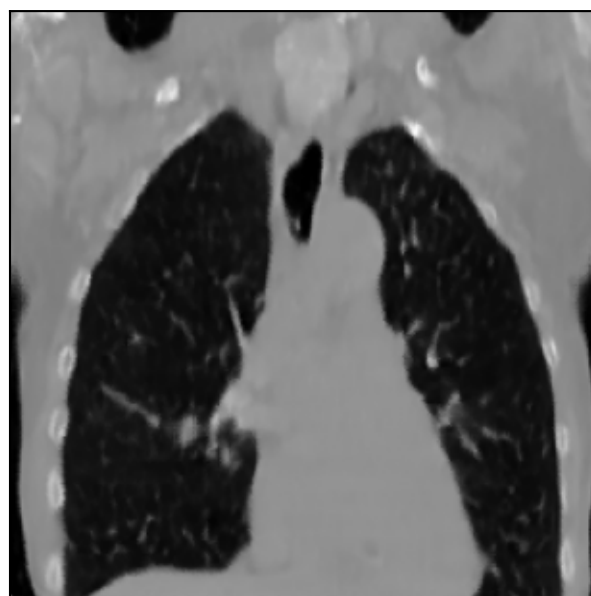
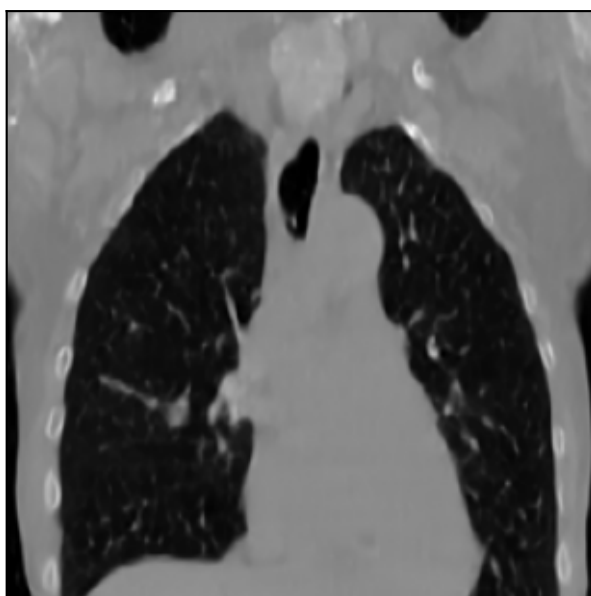
Warped (50→10)

|Fixed - Warped|

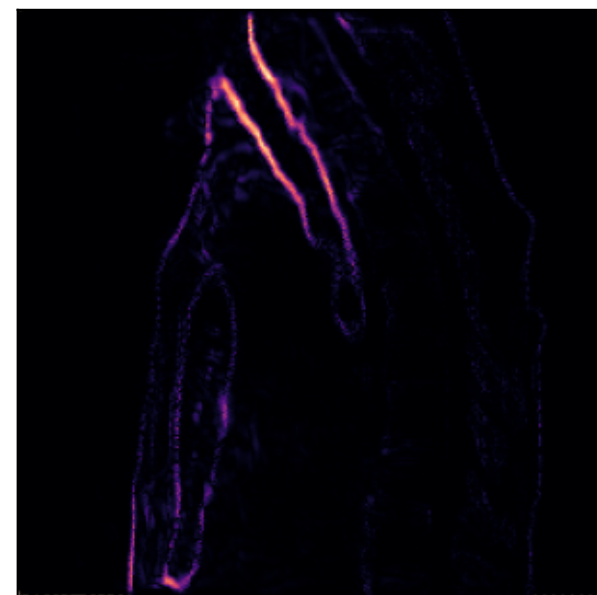
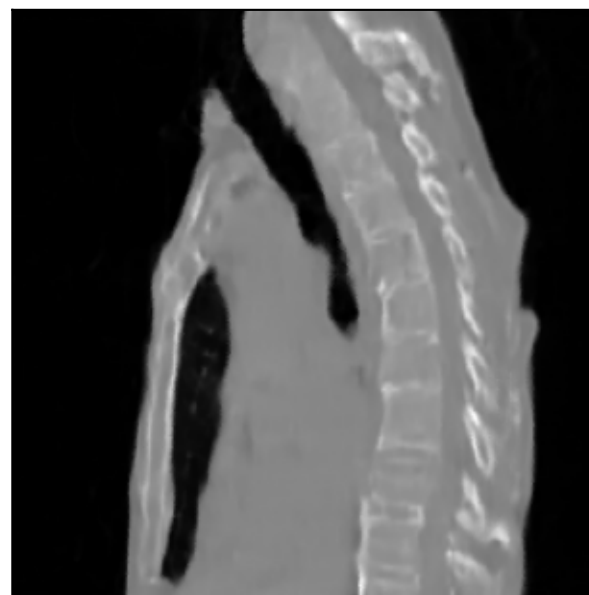
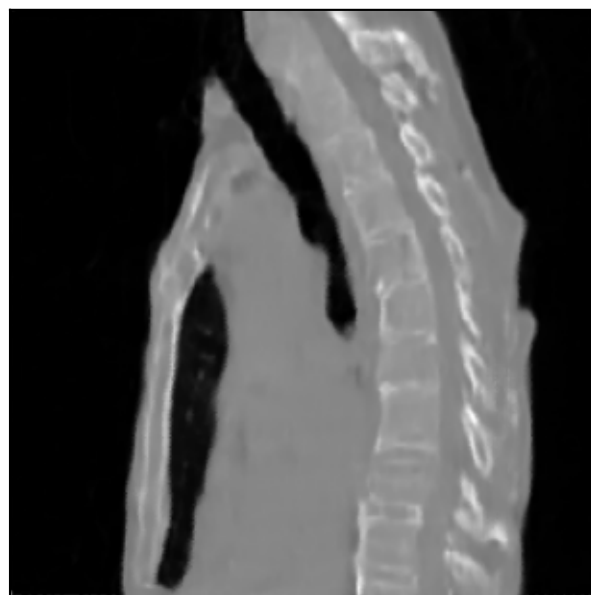
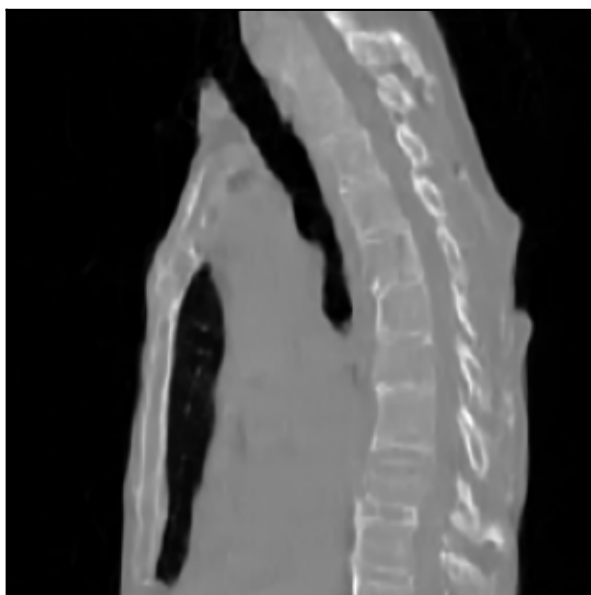
Axial mid



Coronal mid

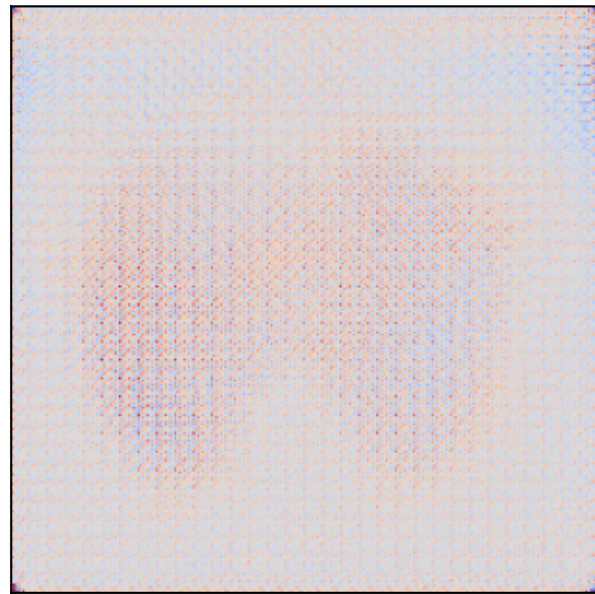


Sagittal mid

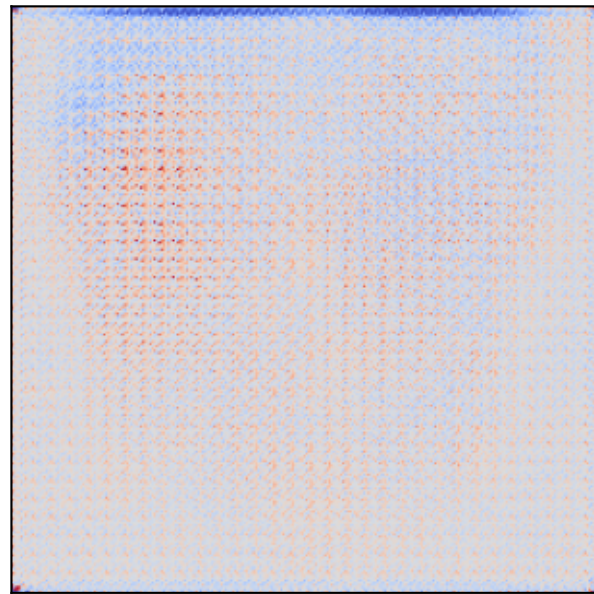


Jacobian determinant + histogram

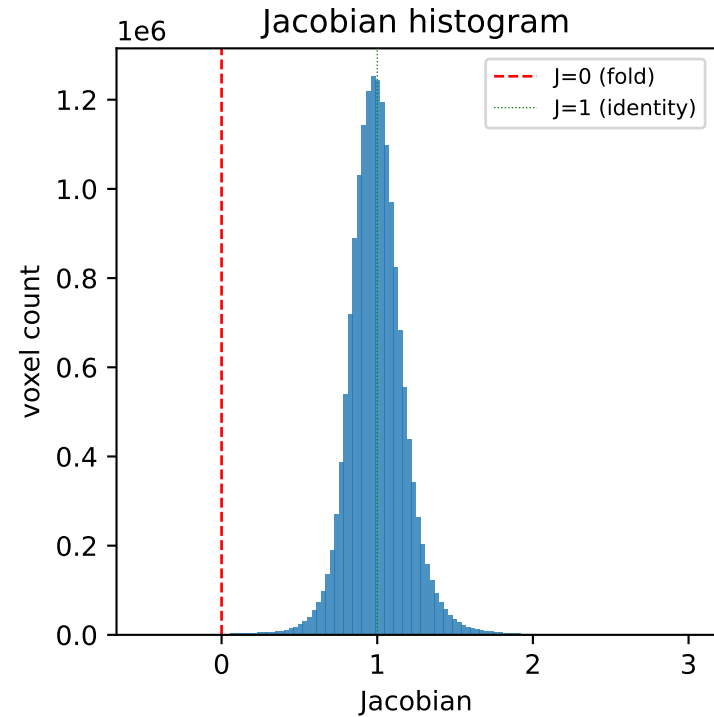
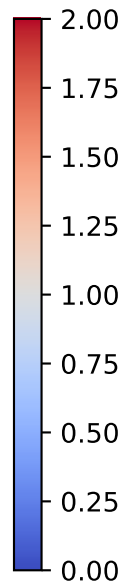
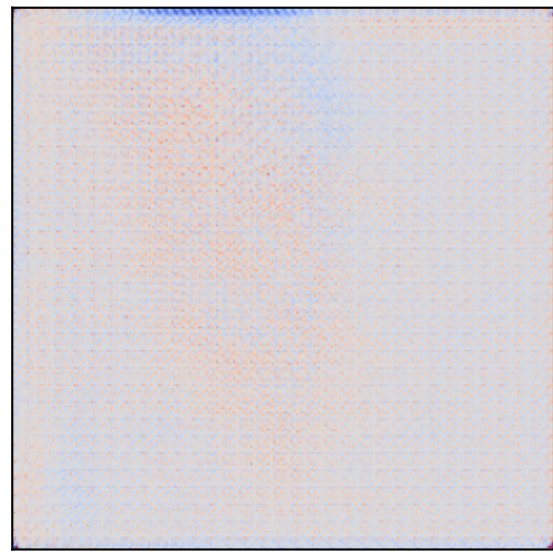
Jacobian Axial mid



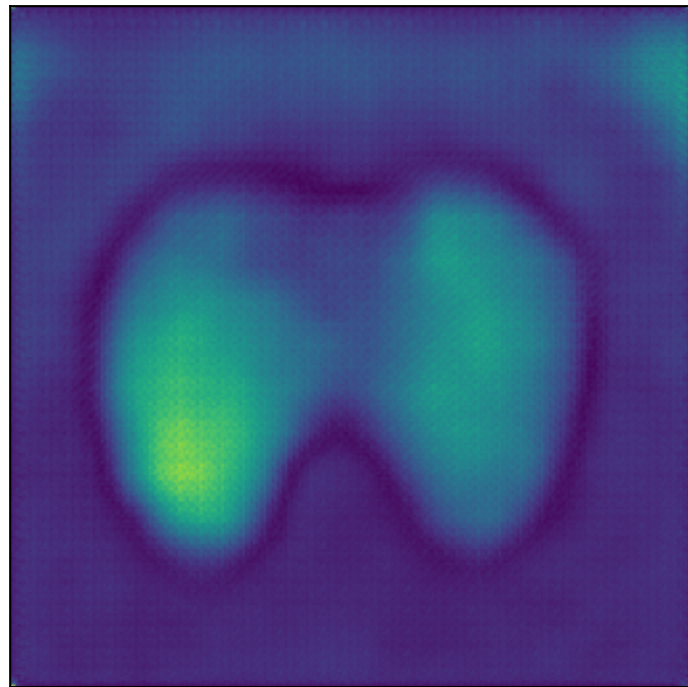
Jacobian Coronal mid



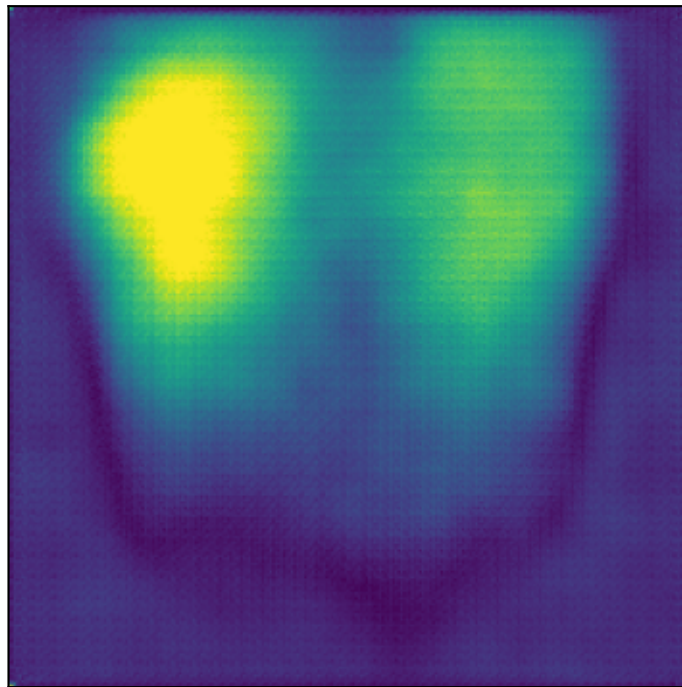
Jacobian Sagittal mid



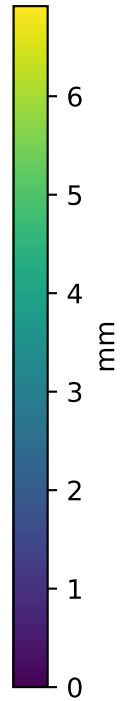
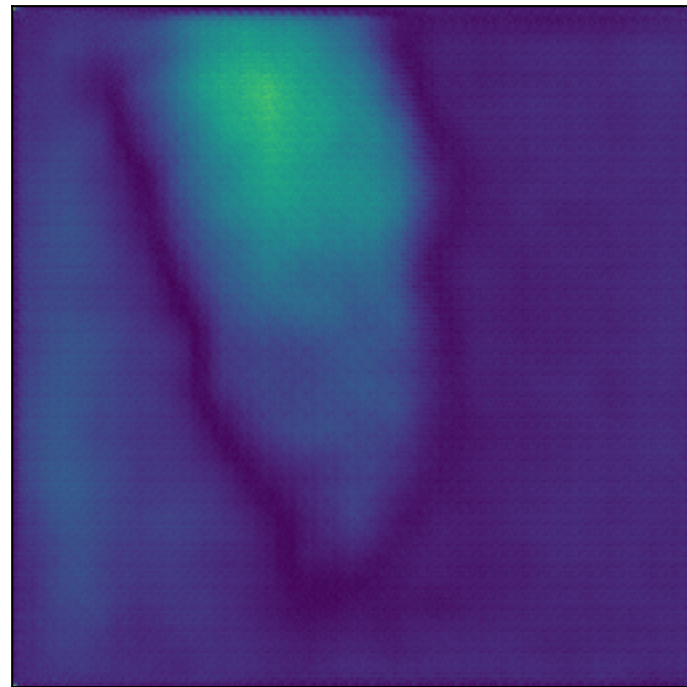
$|u|$  Axial mid



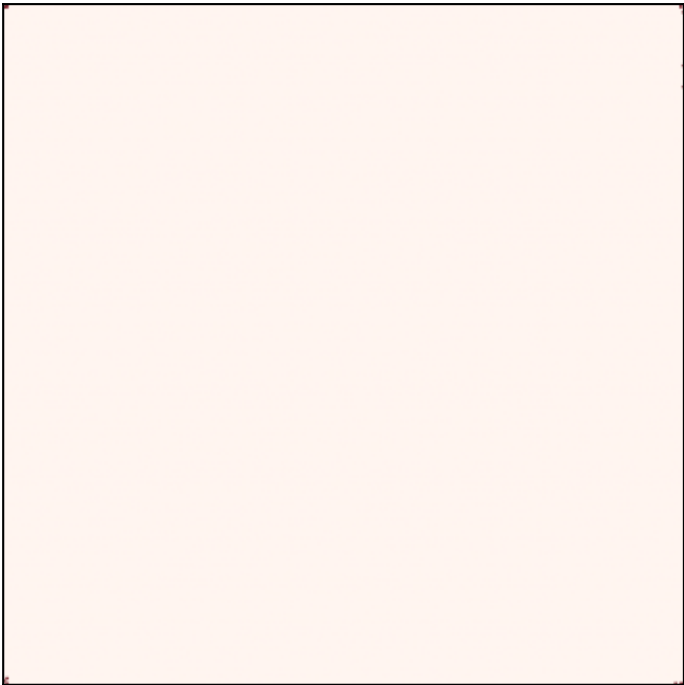
DVF magnitude  
 $|u|$  Coronal mid



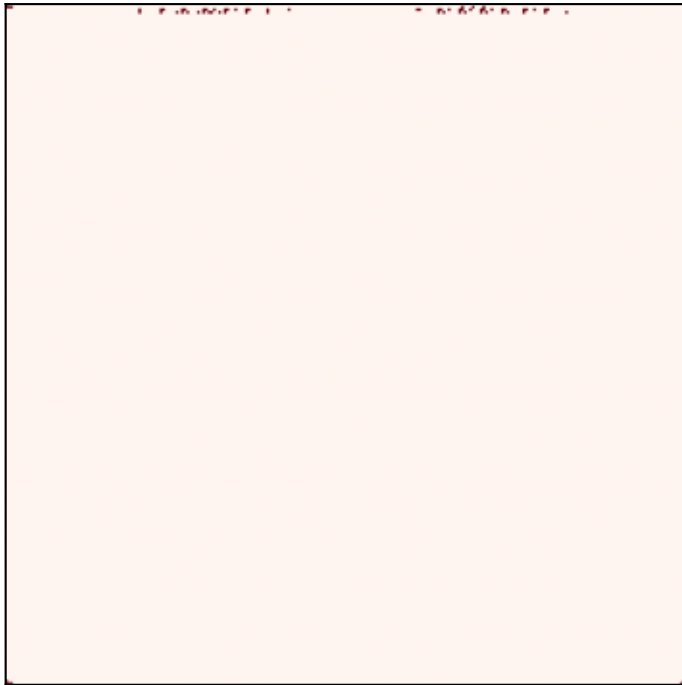
$|u|$  Sagittal mid



Folding mask Axial mid



Folding voxels ( $J \leq 0$ )  
Folding mask Coronal mid



Folding mask Sagittal mid

